Qizhen Lan

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Research Interests

- My research spans agentic AI, medical world models, and efficient visual perception. I focus on evidence-grounded reasoning, action-aware simulation, and tool-using clinical workflows, with an interest in vision-language-action (VLA) systems for embodied medical AI.

Education

- **University of Alabama at Birmingham** — *Jan 2024 – May 2025*
  Ph.D. in Computer Science — Birmingham, United States
- **Bowling Green State University** — *Aug 2019 – Dec 2023*
  Ph.D. student in Data Science — Bowling Green, United States
- **Bowling Green State University** — *Aug 2018 – Jul 2019*
  Master of Science in Analytics — Bowling Green, United States
- **Bowling Green State University** — *Aug 2014 – Jun 2018*
  Bachelor of Science in Business Analytics and Intelligence — Bowling Green, United States

Research & Project Experience

- **University of Texas Health Science Center at Houston [🌐]** — *May 2025 – present*
  Postdoctoral Research Fellow (mentor: Dr. Xiaoqian Jiang) — Houston, TX, United States
  - **Medical world models:** Co-developed a capability and SATO-V evidence framework for medical world models, distinguishing action-conditioned simulation, counterfactual reasoning, and model-based planning in a survey submitted to TMLR.
  - **Agentic medical AI:** Developed ClinAgent, an agentic medical-imaging research platform that orchestrates brain segmentation, longitudinal analysis, and evidence-linked reporting in a unified workflow for physician review.
  - **Neuroimaging AI infrastructure:** Built complementary resources: BrainTRACE evaluates evidence-grounded MRI reasoning on **7,273 scored questions**, and OpenBrain supplies **35,838 cases** with generated segmentation labels, provenance, and quality scores.
  - **Efficient medical imaging:** Developed compact 3D segmentation through distillation, achieving **93.7% fewer parameters** and **3.0× CPU inference speedup** while retaining **98.7% of teacher Dice** on Mindboggle-101 (npj Health Systems).
  - **Autonomous agentic workflows:** Co-developed execution-guided workflow generation in ComfySearch, surpassing ComfyMind by **7.5 percentage points** (**71.5% task completion** on 200 ComfyBench tasks), alongside difficulty-aware multi-agent orchestration in DAAO.
- **UAB & Bowling Green State University (BGSU)** — *Aug 2021 – May 2025*
  Research Assistant (Advisor: Dr. Qing Tian) — United States
  BGSU: Aug 2021 – Jan 2024 UAB: Jan 2024 – May 2025
  - **Efficient visual perception:** Developed adaptive teacher-student attention for distillation (ACAM-KD), raising RetinaNet-R50 by **3.8 mAP on COCO 2017 val** over the undistilled student without adding inference-time computation.
  - **Transformer detection for driving scenes:** Developed context- and location-aware transformer distillation (CLoCKDistill), boosting DINO-R18 by **6.4 AP points** (56.1 to 62.5 COCO-style AP on KITTI) at unchanged inference cost.

Technical Expertise

- **Modeling:** PyTorch, nnU-Net; distillation, 3D segmentation, transformer detection.
- **Agent systems:** Tool orchestration, execution feedback, model routing; Python, FastAPI, React.
- **Evaluation:** Multimodal benchmarks, longitudinal MRI, dataset provenance, reference-free QC.

SELECTED PUBLICATIONS

J=JOURNAL, C=CONFERENCE, W=WORKSHOP, M=MANUSCRIPT

[M.1] *Medical World Models: From Static Prediction to Clinically Valid Action-Aware Simulation*. **Transactions on Machine Learning Research (TMLR), 2026** (Submitted; survey manuscript).

[M.2] **Qizhen Lan**, Yu-Chun Hsu, Nida Saddaf Khan, Xiaoqian Jiang. *Toward Clinically Deployable 3D Auto-Contouring: A Region- and Context-Aware Knowledge Distillation Framework*. **npj Digital Medicine, 2026** (In preparation).

[J.1] **Qizhen Lan**, Aaron Choi, Jun Ma, Bo Wang, Zhongming Zhao, Xiaoqian Jiang, Yu-Chun Hsu. *From Performance to Practice: Knowledge-Distilled Segmentator for On-Premises Clinical Workflows*. **npj Health Systems, 2026**.

[J.2] Jung Im Choi*, **Qizhen Lan***, Qing Tian. *Boosting Deep Detector Efficiency and Robustness through Detection Discriminant Reorganization and Compression*. **Neural Networks, 2026** (* Co-first author; Accepted).

[J.3] **Qizhen Lan**, Qing Tian. *Instance, Scale, and Teacher Adaptive Knowledge Distillation for Visual Detection in Autonomous Driving*. **IEEE Transactions on Intelligent Vehicles (T-IV), 2022**.

[C.1] Mengchen Fan*, **Qizhen Lan***, Xi Xiao, Lin Zhao, Liang He, Tianyang Wang, Xiaoqian Jiang, Pulin Che, and Baocheng Geng. *MedSTP: Preserving Subspace and Transition Evidence for Medical VQA Token Pruning*. **WACV, 2027** (* Co-first authors).

[C.2] Yuke He, Liang Fan, Yusheng Tan, Yuxiang Wei, Yanteng Zhang, Qishi Zhan, Liang Chen, and **Qizhen Lan†**. *Do Medical VLMs Look at the Lesion? A Lesion-Localized Counterfactual Audit of Chest X-Ray VLMs*. **WACV, 2027** († Sole corresponding author).

[C.3] Liang Chen, Ziang Song, Zijun Zhang, **Qizhen Lan**, Qishi Zhan, and Xinzhu Peng. *The Scope of Not: Benchmarking and Enhancing Vision-Language Models for Fine-Grained Negation Understanding*. **WACV, 2027**.

[C.4] **Qizhen Lan***, Mengchen Fan*, Hang Zhang*, Jingwei Duan, Moule Lin, Jialin Chen, Baocheng Geng, and Xiaoqian Jiang†. *BrainTRACE: Tracing Longitudinal, Multimodal, and Volumetric Evidence in Brain MRI Clinical Reasoning*. **NeurIPS Evaluations & Datasets Track, 2026** (* Co-first authors; † Corresponding author).

[C.5] **Qizhen Lan**, Yu-Chun Hsu, Yuxiang Wei, Lijing Zhu, Zenan Sun, Liang He, Lishan Yu, and Xiaoqian Jiang. *OpenBrain: An Auditable Generated-Label Release for Whole-Brain MRI Parcellation*. **NeurIPS Evaluations & Datasets Track, 2026**.

[C.6] Liang He, SiYuan Ma, Qishi Zhan, Yongqi Fan, Mingyu Cao, **Qizhen Lan**, Zhaolu Kang, and Xilu Wang. *ValuSpec: Plug-and-Play Candidate Valuation before Target Verification for Tree-Based Speculative Decoding*. **NeurIPS, 2026** (Poster).

[C.7] Jinwei Su*, **Qizhen Lan***, ZEYU WANG, Yinghui Xia, Hairu Wen, Yiqun Duan, Xi Xiao, TIANYU SHI, Yang Jingsong, and Lewei He. *ComfySearch: Autonomous Exploration and Reasoning for ComfyUI Workflows*. **EMNLP Findings, 2026** (* Co-first author; Accepted). [Paper]

[C.8] Tingyu Wu, CaoGuangyu, **Qizhen Lan**, Zhisheng Chen, Ziyan Weng, Bingkun Zhu, Miao Su, Chenglong Li, Zhengwei Xie, Huacan Wang, Sen Hu, Haocun Ye, NAN AN, Zaoqu Liu, and Ronghao Chen. *KnowMeBenchV2: Evidence-Grounded Person-Centric Long-Video Understanding*. **EMNLP Main Conference, 2026** (**Oral Presentation**).

[C.9] Zhisheng Chen, Yingwei Zhang, **Qizhen Lan**, Tianyu Liu, Huacan Wang, Yi Ding, Ziyu Jia, Ronghao Chen, Kun Wang, and Xinliang Zhou. *Uni-NTFM: A Unified Foundation Model for EEG Signal Representation Learning*. **ICLR, 2026**.

[C.10] Jinwei Su*, **Qizhen Lan***, Yinghui Xia, Lifan Sun, Weiyou Tian, Tianyu Shi, Xinyuan Song, and Lewei He. *Difficulty-Aware Agentic Orchestration for Query-Specific Multi-Agent Workflows*. **WWW, 2026** (* Co-first author).

[C.11] Tingyu Wu, Zhisheng Chen, Ziyan Weng, Shuhe Wang, Shuo Zhang, Sen Hu, Silin Wu, **Qizhen Lan†**, Huacan Wang, and Ronghao Chen. *KnowMe-Bench: Benchmarking Person Understanding for Lifelong Digital Companions*. **ACL Main Conference, 2026** († Corresponding author; Oral Presentation).

[C.12] Xi Xiao, Chen Liu, Chih-Ting Liao, Yunbei Zhang, **Qizhen Lan**, Yuxiang Wei, Lin Zhao, Janet Wang, Jianyang Gu, Muchao Ye, Tianyang Wang, and Hao Xu. *Staying VIGILant: Mitigating Visual Laziness via Counterfactual Visual Alignment in MLLMs*. **ECCV, 2026**.

[C.13] Mengchen Fan, Baocheng Geng, Xi Xiao, Tianyang Wang, Siyuan Mei, Pulin Che, Xiaoqian Jiang†, and **Qizhen Lan†**. *Detail Consistent Stage-Wise Distillation for Efficient 3D MRI Segmentation*. **MICCAI, 2026** († Corresponding authors; **Early Accept**; **Spotlight Presentation**).

[C.14] Zhicheng Ding*, Xinyu Chu*, Jung Im Choi, Qing Tian†, Tianyu Shi, Xiaoqian Jiang, Lijing Zhu, and **Qizhen Lan†**. *Displacement Preserving Relational Distillation for Robust Medical Segmentation*. **MICCAI, 2026** (* Co-first authors; † Corresponding authors). [Code]

[C.15] Siyuan Mei, Yanteng Zhang, Yan Xia, **Qizhen Lan**, Yipeng Sun, Siming Bayer, Zirong Li, Chengze Ye, Daiqi Liu, Xiaoqian Jiang, Fuxin Fan, Yixing Huang, and Andreas Maier. *Time Matters: Rethinking Diffusion and Flow Models in One-Step Medical Image Translation*. **MICCAI, 2026** (**Early Accept**; **Oral Presentation**; **Nominated and Shortlisted for the Best Paper and Young Scientist Award**).

[C.16] **Qizhen Lan**, Qing Tian. *CLoCKDistill: Consistent Location-and-Context-aware Knowledge Distillation for DETRs*. **WACV, 2026**.

[C.17] **Qizhen Lan**, Jong-Im Choi, Qing Tian. *Visual Detector Compression via Location-Aware Discriminant Analysis*. **WACV, 2026**.

[C.18] Chen Liu*, **Qizhen Lan***, Zhicheng Ding, Xinyu Chu, Qing Tian. *Learnable Instance Attention Filtering for Adaptive Detector Distillation*. **ICASSP, 2026** (* Co-first author).

[C.19] Ting Huang, Biao Wu, Ronghao Chen, Zeyu Zhang, Tengfei Cheng, **Qizhen Lan**, Huacan Wang, and Hao Tang. *DramaAgent: Agentic Storytelling Video Generation*. **AACL-IJCNLP Main Conference, 2026**. [Paper]

[C.20] **Qizhen Lan**, Qing Tian. *ACAM-KD: Adaptive and Cooperative Attention Masking for Knowledge Distillation*. **ICCV, 2025**.

[C.21] Lijing Zhu, **Qizhen Lan**, Qing Tian, Wenbo Sun, Li Yang, Lu Xia, Yixin Xie, Xi Xiao, Tiehang Duan, Cui Tao, Shuteng Niu. *ETT-CKGE: Efficient Task-driven Tokens for Continual Knowledge Graph Embedding*. **ECML-PKDD, 2025**.

[C.22] Yanteng Zhang, Songheng Li, Zeyu Shen, **Qizhen Lan**, Lipei Zhang, Yang Liu, Vince Calhoun. *Lost in Distortion: Uncovering the Domain Gap Between Computer Vision and Brain Imaging–A Study on Pretraining for Age Prediction*. **ISBI, 2025**.

[C.23] Jong-Im Choi*, **Qizhen Lan***, Qing Tian. *Improving Deep Detector Robustness via Detection-Related Discriminant Maximization and Reorganization*. **WACV, 2025** (Oral; * Co-first author).

[C.24] Zhicheng Ding, **Qizhen Lan**, Qing Tian. *Target-Driven and Student-Centered Knowledge Distillation for Traffic Object Tracking*. **IV, 2025**.

[C.25] **Qizhen Lan**, Qing Tian. *Gradient-Guided Knowledge Distillation for Object Detectors*. **WACV, 2024**.

[C.26] Wentao Wang, Xi Xiao, Mingjie Liu, **Qizhen Lan**, Xuanyao Huang, Qing Tian, Swalpa Kumar Roy, Tianyang Wang. *Multi-dimension Transformer with Attention-based Filtering for Medical Image Segmentation*. **ICTAI, 2024**.

[W.1] Yuxi Lu, Kunqi Li, Zhidong Li, Chenya Huang, Biao Wu, **Qizhen Lan**, Ronghao Chen, and Bin Liang. *Interventional Analysis of Evidence Faithfulness in Generative Cloud Removal*. **NeurIPS XAI4Science Workshop, 2026** (Non-archival).

Honors and Awards

- **NSF ACCESS Advanced Computing Allocation (≈$10,000 GPU resources)** *2026*
- **Graduate College Professional Development Funds** *2023*
- **Analytics Instructional Fee Award** *2018*

Academic Services

- **Workshop Organizer**
  - 1st Latent Visual Reasoning (LVR) Workshop, WACV 2027
- **Conference Reviewer**
  - Annual Conference on Neural Information Processing Systems (NeurIPS), 2026
  - ACM International Conference on Multimedia (ACM MM), 2026
  - ACL Rolling Review (ARR), 2026
  - International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2026
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
  - Association for the Advancement of Artificial Intelligence (AAAI), 2025, 2026
  - International Symposium on Biomedical Imaging (ISBI), 2025, 2026
  - IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024–2026
- **Journal Reviewer**
  - IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
  - Nature - npj Digital Medicine
  - IEEE Transactions on Multimedia (TMM)
  - IEEE Transactions on Neural Networks and Learning Systems (TNNLS)